Week 21: The shortest route that touches a line

Adult guide built from the problem files · Unpiloted; string handling and the fold are unrehearsed.

This week's adult guide is week-21-facilitator.pdf, built from editable/facilitator-src/. These problem files hold each problem's one-sentence idea; the full keys, proofs and hints are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Two routes with the same length	P1			first
Shorter than both drawn routes	P2			first
Which shortest route is shorter		P1	P1	first
Reflected finish dots tie	P3	P2	P2	first
Unequal heights and a proof	P4	P3	P3	first
Which shortest route is shorter (K–1)	P5			return
A vertical line	P6	P4		K-1 return; 2-3 first
Two journeys share the best contact			P4	return
Choose the start for a given contact	P7	P5	P5	return
Equal minimum lengths		P6		return
Restricting where a route may touch		P7	P6	return
Can two best contacts exist?			P7	return

First route. K–1 P1–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P3. **Return visits.** K–1 P5–P7; Grades 2–3 P5–P7; Grades 4–5 P4–P7.

Key, one entry per problem

Two routes with the same length

K–1 P1 · first route

Idea. Mirror-image contacts give equal routes when A and B are at the same height.

Shorter than both drawn routes

K–1 P2 · first route

Idea. Both drawn guesses are 0.7 cm or more longer than the best route, so string can beat them.

Which shortest route is shorter

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. C is nearer A, yet its shortest route needs about 1 cm more string than B's.

Reflected finish dots tie

K–1 P3 · Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. C is B reflected in the line, so the two routes through any one contact have the same length.

Unequal heights and a proof

K–1 P4 · Grades 2–3 P3 · Grades 4–5 P3 · first route · wording differs by band

Idea. Folding B to B′ makes every route as long as a bent path from A to B′, so the straight A–B′ crossing is the unique best contact.

Which shortest route is shorter (K–1)

K–1 P5 · return route

Idea. The shortest A–B route needs less string than the shortest A–C route.

A vertical line

K–1 P6 · Grades 2–3 P4 · K-1 return; 2-3 first route

Idea. The same fold works on a vertical line for each of the three pairs.

Two journeys share the best contact

Grades 4–5 P4 · return route

Idea. A–B and C–D both cross the line at 6.2 cm: different journeys can share a best contact.

Choose the start for a given contact

K–1 P7 · Grades 2–3 P5 · Grades 4–5 P5 · return route · wording differs by band

Idea. The starts whose best contact to B is P are exactly the ray from P away from B′.

Equal minimum lengths

Grades 2–3 P6 · return route

Idea. Both shortest routes are $\sqrt{164}$ cm long, though they touch at different points.

Restricting where a route may touch

Grades 2–3 P7 · Grades 4–5 P6 · return route · wording differs by band

Idea. The best contact 6.5 cm is allowed between C and D and between C and F, so it stays shortest there; between E and F it is not allowed.

Can two best contacts exist?

Grades 4–5 P7 · return route

Idea. Equality in the triangle inequality happens only on A–B′, so the best contact is unique.